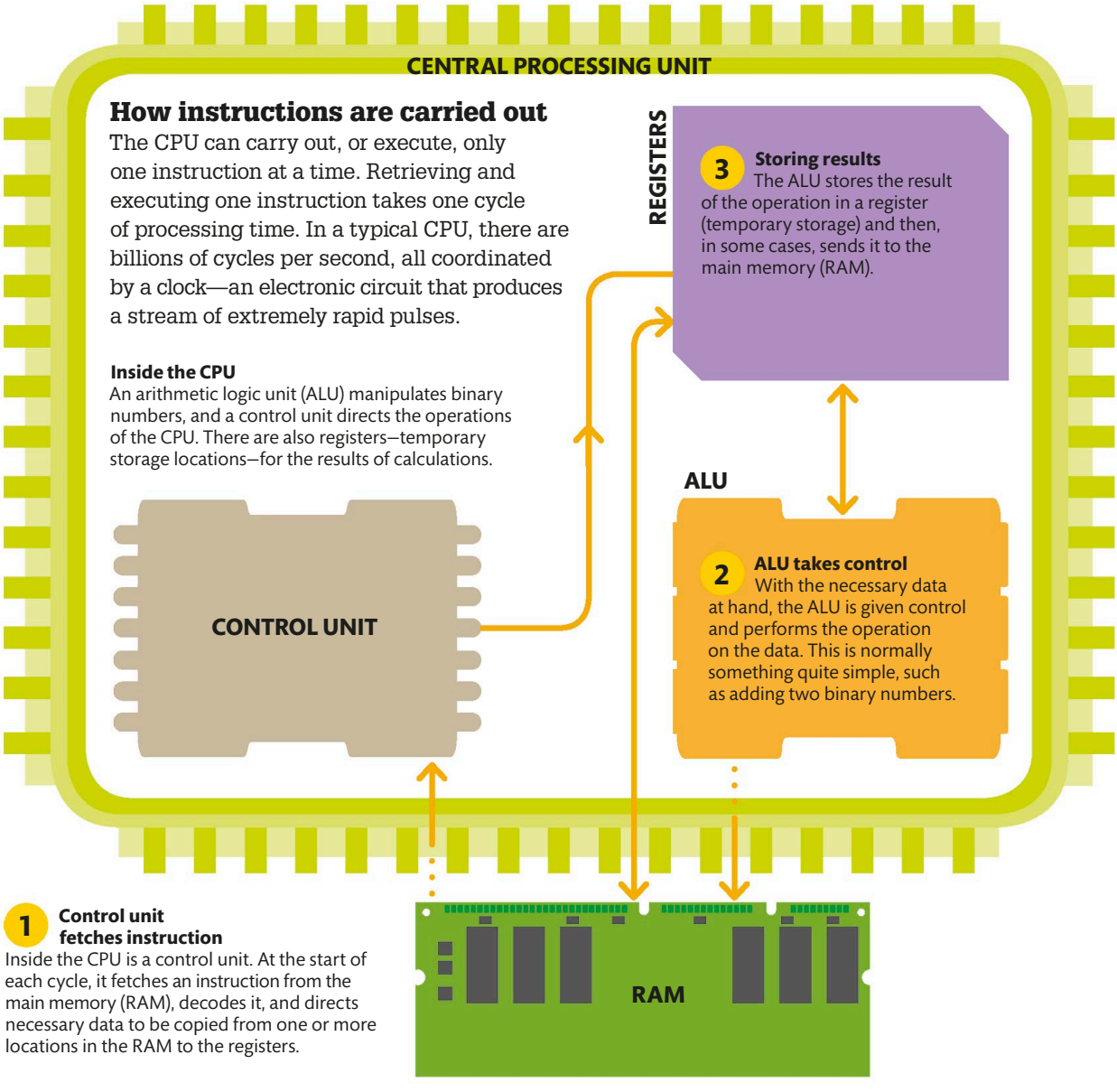
How instructions are carried out

The CPU can carry out, or execute, only one instruction at a time. Retrieving and executing one instruction takes one cycle of processing time. In a typical CPU, there are billions of cycles per second, all coordinated by a clock—an electronic circuit that produces a stream of extremely rapid pulses.

Inside the CPU
An arithmetic logic unit (ALU) manipulates binary numbers, and a control unit directs the operations of the CPU. There are also registers—temporary storage locations—for the results of calculations.



1 Control unit fetches instruction
Inside the CPU is a control unit. At the start of each cycle, it fetches an instruction from the main memory (RAM), decodes it, and directs necessary data to be copied from one or more locations in the RAM to the registers.

MACHINE CODE

The data and instructions the CPU manipulates arrive as a stream of binary numbers—1s and 0s. This stream of numbers is called machine code and is broken into chunks, typically 32 or 64 binary digits (bits) long.

```
0 1 1 0 0 1 0 1
0 0 1 1 0 1 0 0
0 0 1 0 1 1 1 0
1 0 0 1 0 1 0 0
```

THE WORLD'S SMALLEST COMPUTER IS SMALLER THAN A GRAIN OF SALT

